
Indian Institute of Technology Palakkad
Algorithms
Assignment: Binary Search Trees

1. Show that if a node in a binary search tree has two children, then its successor has no left child and its predecessor has no right child.
2. Consider a binary search tree T whose keys are distinct. Show that if the right subtree of a node x in T is empty and has a successor y , then y is the lowest ancestor of x whose left child is also an ancestor of x . Recall that every node is its own ancestor.
3. An alternative method of performing an inorder tree walk of an n -node binary search tree finds a minimum element in the tree and then makes $n - 1$ calls to finding a successor. Prove that this algorithm runs in $\mathcal{O}(n)$ time.
4. Let T be a binary search tree whose keys are distinct, let x be a leaf node, and let y be its parent. Show that $y.key$ is either the smallest key in T larger than $x.key$ or the largest key in T smaller than $x.key$.
5. Is the operation of deletion commutative in the sense that deleting x and then y from a binary search tree leaves the same tree as deleting y and then x ? Justify your answer.
6. Equal keys pose a problem for the implementation of binary search trees. What is the asymptotic performance of INSERT when used to insert items with identical keys into an initially empty binary search tree?

```

TREE-INSERT( $T, z$ )
1   $x = T.root$            // node being compared with  $z$ 
2   $y = NIL$               //  $y$  will be parent of  $z$ 
3  while  $x \neq NIL$        // descend until reaching a leaf
4       $y = x$ 
5      if  $z.key < x.key$ 
6           $x = x.left$ 
7      else  $x = x.right$ 
8   $z.p = y$               // found the location—insert  $z$  with parent  $y$ 
9  if  $y == NIL$ 
10      $T.root = z$         // tree  $T$  was empty
11 elseif  $z.key < y.key$ 
12      $y.left = z$ 
13 else  $y.right = z$ 
```

Consider changing INSERT to test whether $z.key = x.key$ before line 5 and to test whether $z.key = y.key$ before line 11. If equality holds, implement one of the following strategies. For each strategy, find the asymptotic performance of inserting items with identical keys into an initially empty binary search tree. The strategies are described for line 5, which compares the keys of z and x . Substitute y for x to arrive at the strategies for line 11.

- (a) Keep a boolean flag $x.b$ at node x , and set x to either $x.left$ or $x.right$ based on the value of $x.b$, which alternates between FALSE and TRUE each time INSERT visits x while inserting a node with the same key as x .
- (b) Keep a list of nodes with equal keys at x , and insert z into the list.